

ADVAIT PARULEKAR

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EDUCATION

PhD, Dept. of Electrical and Computer Engineering, UT Austin, January 2020-Present

- Focus on Learning Theory - LLMs, Representation Learning, Online Learning.

BS Computer Engineering, Texas A&M University, August 2015-May 2019

- Graduated summa cum laude with departmental honors, minor in Mathematics

WORK

Research Intern - Data and Machine Learning Innovation

Apple

Cupertino, USA (May 2025 - Sept 2025)

- Zero-Shot "Out of Distribution Detection" for on-device CLIP models.

Research Intern - Machine Learning and Optimization team

Google Research India

Bengaluru, India (June 2023 - Sept 2023)

- Studied simplicity bias in wide shallow neural networks.

PUBLICATIONS

In-Context Learning with Transformers: Softmax Attention Adapts to Function Lipschitzness.

Advait Parulekar*, Liam Collins*, Aryan Mokhtari, Sujay Sanghavi, Sanjay Shakkottai. (*co-first authors*)

Spotlight @ Conference on Neural Information Processing Systems (**NeurIPS 2024**)

InfoNCE Loss Provably Learns Cluster-Preserving Representations.

Advait Parulekar, Liam Collins, Karthikeyan Shanmugam, Aryan Mokhtari, Sanjay Shakkottai.

Conference on Learning Theory (**COLT 2023**), Bengaluru, India, July 2023.

PAC Generalization via Invariant Representations.

Advait Parulekar, Karthikeyan Shanmugam, Sanjay Shakkottai.

Proceedings of the 40th International Conference on Machine Learning (**ICML 2023**), Honolulu, HI, July 2023.

Regret Bounds for Stochastic Shortest Path Problems with Linear Function Approximation.

Daniel Vial, **Advait Parulekar**, Sanjay Shakkottai and R. Srikant.

Proceedings of the 39th International Conference on Machine Learning (**ICML 2022**), Baltimore, MD, July 2022.

Improved Algorithms for Misspecified Linear Markov Decision Processes.

Daniel Vial, **Advait Parulekar**, Sanjay Shakkottai and R. Srikant.

Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (**AISTATS 2022**), Virtual Conference, April 2022.

L1 Regression with Lewis Weights Subsampling.

Aditya Parulekar, **Advait Parulekar**, Eric Price.

The International Conference on Randomization and Computation (**RANDOM 2021**), Virtual Conference, August 2021

A quadratically convergent iterative scheme for locating conical degeneracies in the spectra of parametric self-adjoint matrices.

Gregory Berkolaiko, **Advait Parulekar**. (*alphabetical order*)

SIAM Journal on Matrix Analysis and Applications, 2021, Vol. 42, No. 1 : pp. 224-242.

PREPRINTS

Efficient Approximate Posterior Sampling with Annealed Langevin Monte Carlo.

Advait Parulekar, Litu Rout, Karthikeyan Shanmugam, Sanjay Shakkottai.

<https://arxiv.org/abs/2508.07631>

Under review ICLR 2026.

A Theoretical Justification for Image Inpainting using Denoising Diffusion Probabilistic Models.

Litu Rout, **Advait Parulekar**, Constantine Caramanis, Sanjay Shakkottai.

<https://arxiv.org/abs/2302.01217>

Stochastic Linear Bandits with Protected Subspace. **Advait Parulekar**, Soumya Basu, Aditya Gopalan, Karthikeyan Shanmugam, Sanjay Shakkottai. <https://arxiv.org/abs/2011.01016>

COMPETITIONS

2021 NSF Graduate Research Fellowship Program competition, Honorable Mention

2018 William Lowell Putnam Math Contest, Honorable Mention

2015 USA Physics Olympiad Team Member

2013 Indian Math Olympiad Training Camp

OTHER

2022 IPhO Grader

Zurich, Switzerland (July 2022)

- Graded the theory solutions from a selection of countries for the International Physics Olympiad.

Research Assistant

Dept. of Mathematics, advisor: Prof. Gregory Berkolaiko

Texas A&M University (Aug 2019-Dec 2019)

- Spectral Theory

Curriculum Development

Art of Problem Solving

(Aug 2019)

- Writing handouts for PhysicsWOOT and grader for PhysicsWOOT.

SMaRT Camp Counsellor

Dept. of Mathematics

Texas A&M University (Summer 2017, Summer 2018, Summer 2019)

- Helped teach number theory, modern algebra, linear algebra, to high school students.
- <https://github.com/advaitparulekar/Inv-Radon-Transform>

Peer Teacher (Undergraduate TA)

Dept. of Electrical Engineering & Dept. of Computer Science

Texas A&M University (Spring 2019)

- “Introduction to Computer Systems,” “Structured Programming in C,” “Programming Studio,” “Signals and Systems,” “Electronics,” “Electrical Circuit Theory,” and “Random Signals and Systems.”

GRADUATE COURSEWORK

Theory of Computation, Online Learning, Methods in Applied Mathematics (Functional Analysis), Stochastic Geometry, Stochastic Approximation, Advanced Probability, Sub-linear Algorithms, Combinatorial Optimization, Markov Chains and Mixing Times, Convex Optimization, Statistical Machine Learning, Game Theory, Information Theory, Cryptography, Randomized Algorithms, Algorithms in Structural Bio-informatics.

REVIEWING

AISTATS '21, AISTATS '22, ICML '22, RANDOM '22, NeurIPS '22